Basin Shaping for Opponent-Aware Learning: Local Meta-MAPG Convergence with Restarts

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Abstract

Multiagent reinforcement learning is difficult because each agent experiences the others as a moving target. Meta-MAPG addresses this non-stationarity by differentiating through both an agent's own future updates and the future updates of its peers, but its local stochastic-approximation status has been unclear. We study Meta-MAPG from a local convergence viewpoint. Our analysis treats the finite-unroll Meta-MAPG correction as an opponent-aware shaping field added to the base game-gradient dynamics. In phase 1, constant shaping is used only as a basin-entry mechanism: if the shaped field improves local drift by $\lambda\mu_M$, the certified entry time shrinks by the same factor. In phase 2, the shaping coefficient is annealed to zero, so the limiting ODE returns to the original game-gradient flow and Giannou et al.'s stable-Nash theorem applies. Independent restarts then globalise the local guarantee with an almost-sure finite-restart bound and a geometric tail. Experiments in Stag Hunt and iterated prisoner's dilemma match this picture: peer-learning terms drive the empirical gain, restart budget becomes equilibrium-selection budget, basin maps expand under opponent-aware shaping, and a small MLP IPD check shows the effect is not purely tabular. During phase 1 the dynamics are centered at a shifted fixed point $\phi^*_\lambda = \phi^* + O(\lambda)$, and annealing removes this shift so the asymptotic target returns to the original Nash equilibrium ϕ^*.

1. Introduction

Learning in stochastic games is hard because every learner changes the environment faced by the others. From the perspective of any single agent, the environment is therefore effectively non-stationary. Giannou et al. [5] prove local convergence of policy gradients to stable Nash policies under strategic stability and second-order sufficiency. This is the right baseline for policy-gradient dynamics near a stable Nash point, but it still treats opponent learning as background variation rather than something the update itself can model.

Meta-learning suggests a richer view. Kim et al. [7] model multiagent learning as a Markov chain of joint policy updates and derive Meta-MAPG, whose meta-gradient contains a current-policy term, an own-learning term, and a peer-learning term. This decomposition captures the fact that an agent's current behaviour changes both its own future policy and the future policies of the peers it is

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interacting with. Empirically, those extra terms matter. The open question is when they fit the local stochastic-approximation picture used by the standard convergence theory.

Our paper takes that local question as primary. We treat the finite-unroll Meta-MAPG correction as an opponent-aware shaping field added to the base game-gradient dynamics. The central idea is to separate two roles that are often conflated. Constant shaping can be useful for finite-time basin entry and equilibrium selection, but asymptotic convergence should still target the stable Nash equilibria of the original game. We therefore analyze a two-phase schedule: a constant-shaping phase for basin entry, followed by an annealed phase that turns the limiting ODE back into the original game-gradient flow.

Our contributions.

1) **Meta-MAPG as local stochastic approximation.** Under bounded rewards, smooth policies, controlled unroll length, and standard sampling assumptions, we show that the finite-unroll Meta-MAPG update can be written as the base game-gradient field plus controlled opponent-aware terms with martingale noise and summable bias.

2) **A two-phase local convergence picture.** We show that constant opponent-aware shaping can improve local drift and certified basin geometry around a desirable equilibrium, while an annealed second phase restores the original game's stable Nash target.

3) **Restart-based globalisation.** We turn the local theorem into a global search statement: with positive restart mass on the basin, independent restarts yield an almost-sure finite-restart guarantee and a geometric tail bound. Restart budget becomes equilibrium-selection budget.

4) **Theory-facing experiments.** Stag Hunt and iterated prisoner's dilemma test the mechanism predicted by the theory: peer-learning changes equilibrium selection, basin maps expand under shaping, and the two-phase schedule preserves the intended target while improving basin entry.

2. From Stochastic Games to Meta-MAPG

2.1. Stochastic Games

We consider finite discounted stochastic games

$$\mathcal{G} = (N, \mathcal{S}, \{\mathcal{A}_i\}_{i\in N}, P, \{r_i\}_{i\in N}, \rho, \gamma_{\text{disc}}),$$

with finite state space $\mathcal{S}$, finite action sets $\mathcal{A}_i$, transition kernel P, bounded rewards $|r_i| \le R_{\max}$, initial-state distribution ρ, and discount factor $\gamma_{\text{disc}} \in [0, 1)$. Each agent has a differentiable stationary Markov policy $\pi_{\phi_i}(a_i \mid s)$ with parameter $\phi_i \in \mathbb{R}^{d_i}$; the joint parameter is $\phi = (\phi_i)_{i=1}^N \in \mathbb{R}^d$. The value of agent i is

$$J_i(\phi) = \mathbb{E}_{\rho,\pi_\phi}\left[\sum_{t\ge 0} \gamma_{\text{disc}}^t r_i(s_t, a_t)\right],$$

and the standard game-gradient field is

$$v(\phi) = \big(\nabla_{\phi_i} J_i(\phi)\big)_{i=1}^N.$$

First-order stationary points of this field correspond to Nash policies under the gradient-dominance property used by Giannou et al. [5]; finite stochastic games possess stationary Nash equilibria [2].

Definition 1 (Stable Nash and SOS) *A Nash parameter ϕ^* is stable if*

$$\langle v(\phi), \phi - \phi^* \rangle < 0$$

for all $\phi \neq \phi^*$ sufficiently close to ϕ^*. It is second-order stationary (SOS) if there exist $\mu > 0$ and $\rho > 0$ such that

$$\langle v(\phi), \phi - \phi^* \rangle \leq -\mu \|\phi - \phi^*\|^2$$

whenever $\|\phi - \phi^*\| \leq \rho$.

The SOS condition is the quantitative stability condition that yields an attracting neighbourhood and a rate in Giannou et al. [5]. We use it as the local anchor.

2.2. A Markov Chain of Joint Policies

Following Kim et al. [7], we view learning as a Markov chain of joint policy parameters $\phi_0, \ldots, \phi_L$, where each inner transition is produced by a policy-gradient update. For agent i, Meta-MAPG decomposes the meta-gradient into

$$G_i^{\mathrm{MAPG}} = G_i^{\mathrm{cur}} + G_i^{\mathrm{own}} + G_i^{\mathrm{peer}}.$$

The current term is the usual policy-gradient score term. The own-learning term differentiates through how ϕ_i^0 affects agent i's adapted parameters $\phi_i^{1:L}$. The peer-learning term differentiates through how ϕ_i^0 affects other agents' future parameters $\phi_{-i}^{1:L}$. The peer term is the Meta-MAPG analogue of LOLA-style opponent shaping; the own term is absent from pure LOLA.

We write the population opponent-aware field as

$$F_L(\phi) = v(\phi) + M_L^{\mathrm{own}}(\phi) + M_L^{\mathrm{peer}}(\phi),$$

where L is the unroll length. In the convergence theorem we use the annealed update

$$\phi_{n+1} = \mathrm{Proj}_{\mathcal{K}}\left[\phi_n + \alpha_n \left(\hat{v}_n + \lambda_n \hat{M}_{L_n,n}\right)\right], \tag{1}$$

Here $\mathcal{K} \subset \mathbb{R}^d$ is compact, $\hat{M}_{L_n,n}$ estimates the combined own/peer correction, and λ_n controls the asymptotic shaping strength. Projection can be replaced by any standard boundedness mechanism. We state it explicitly so the projection lemma of Kushner and Clark [8] and the ODE tracking argument of Borkar [1] apply without extra tightness work.

Definition 2 (Nash-consistent shaping) *Let U be a neighbourhood of a stable Nash set $\mathcal{N}$ of the base game-gradient field v. A scheduled opponent-aware correction $\lambda_n M_{L_n}$ is Nash-consistent on U if its contribution to the stochastic-approximation recursion is either summable,*

$$\sup_{\phi \in U} \|\lambda_n M_{L_n}(\phi)\| \leq \beta_n, \qquad \sum_n \alpha_n \beta_n < \infty,$$

so the limiting ODE on U is $\dot{\phi} = v(\phi)$, or else its non-vanishing limit field $v + M$ has exactly the same isolated SOS equilibria in U as v with the same local attracting neighbourhoods. The convergence theorem uses the first, annealed case; the second case only records when constant shaping is locally equilibrium-preserving.

3. Local Assumptions

The assumptions separate standard stochastic approximation from the Meta-MAPG estimator step.

Assumption 1 (Step sizes and bounded iterates) *The step sizes satisfy $\alpha_n > 0$, $\sum_n \alpha_n = \infty$, and $\sum_n \alpha_n^2 < \infty$. Iterates remain in a compact set $\mathcal{K}$, either by projection in (1) or by an equivalent stability mechanism.*

Assumption 2 (Estimator decomposition) *There is a locally Lipschitz effective field F—equal to v in the annealed case and to an equilibrium-preserving limit $v + M$ in the non-vanishing case (Theorem 2)—such that the sampled update $g_n = \hat{v}_n + \lambda_n \hat{M}_{L_n,n}$ admits the decomposition*

$$g_n = F(\phi_n) + \xi_{n+1} + b_n,$$

where $\{\xi_{n+1}\}$ is a martingale-difference sequence adapted to $\mathcal{F}_n$ with

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0, \qquad \mathbb{E}[\|\xi_{n+1}\|^2 \mid \mathcal{F}_n] \le \sigma^2,$$

and $\|b_n\| \le C\beta_n$ for a deterministic sequence $\beta_n \ge 0$.

Assumption 3 (Truncation and modeling bias) *The unroll schedule L_n and opponent-modeling procedure produce a residual finite-unroll/modeling error absorbed into the bias sequence β_n of Assumption 2.*

(a) Phase 1 (bounded). *Over any finite constant-shaping window, $\beta_n \le \beta_0$ is uniformly bounded.*

(b) Phase 2 (vanishing). *Under the annealed schedule of Theorem 7, the unroll length grows so that $\beta_n \to 0$ and $\sum_n \alpha_n \beta_n < \infty$. The same decay is assumed for the opponent-modeling residual: if $\widetilde{\beta}_n$ denotes the model-mismatch component of the bias, then $\widetilde{\beta}_n = O(\beta_n)$ and $\sum_n \alpha_n \widetilde{\beta}_n < \infty$. A sufficient choice is $L_n = \lceil r \log n / \log(1/\gamma_{\text{disc}}) \rceil$ for any $r > 1 - p$, giving $\beta_n = O(n^{-r})$.*

Assumption 4 (Summability and annealing) *Step sizes, bias, and shaping coefficients jointly satisfy*

$$\sum_n \alpha_n \beta_n < \infty, \qquad \sum_n \alpha_n \lambda_n < \infty,$$

with M_{L_n} uniformly bounded on $\mathcal{K}$. Under these conditions the limiting ODE of (1) is the base game-gradient flow $\dot{\phi} = v(\phi)$, so the opponent-aware correction is asymptotically transient. Phase 1 of Theorem 7 relaxes the $\sum_n \alpha_n \lambda_n < \infty$ requirement to a constant $\lambda_n = \lambda_c$ over a finite-time window only; phase 2 restores it.

Assumption 5 (Strategic stability) *The stable set $\mathcal{N}$ consists of isolated SOS Nash equilibria satisfying Theorem 1. In the non-annealed local geometry statement, the shaped field $F = v + \lambda M$ satisfies the analogous SOS condition around the same equilibrium.*

Remark 3 *The convergence target is the original game's stable Nash set $\mathcal{N}$. If shaping is kept constant, meta-objectives can have non-Nash fixed points. The theorem is therefore annealed, or explicitly equilibrium-preserving. Constant shaping is used for finite-time basin geometry in Theorem 6, not to claim that arbitrary meta-game stationary points are Nash.*

4. Local Meta-MAPG with Restarts

Lemma 4 (Estimator decomposition and noise control) *Under bounded rewards, twice differentiable policies with bounded score and Hessian on $\mathcal{K}$, and finite second moments of the likelihood-ratio estimators, the finite-unroll Meta-MAPG estimator satisfies Assumption 2. In particular, the current-policy, own-learning, and peer-learning components can each be written as a conditional mean plus a martingale-difference error; finite unroll and opponent-modeling truncation contribute the bias b_n.*

Lemma 5 (Local Meta-MAPG convergence) *Suppose Assumptions 1–5 hold and no restarts occur. If ϕ_0 lies in the attracting neighbourhood of an SOS Nash equilibrium $\phi^* \in \mathcal{N}$, then*

$$\phi_n \to \phi^* \qquad \textit{almost surely.}$$

Moreover, under the quantitative bias and variance exponents of Giannou et al. [5], the usual local squared-distance rate is inherited.

Proposition 6 (Local drift improvement at the shifted fixed point) *Fix an SOS Nash equilibrium ϕ^* of v with drift constant $\mu > 0$ and $v, M \in C^1$ near ϕ^*. The own-learning component vanishes at the Nash, $M^{\text{own}}(\phi^*) = 0$, because $\nabla_{\phi_i} J_i(\phi^*) = 0$. The peer-learning component satisfies $M_i^{\text{peer}}(\phi^*) = \eta[\nabla_{\phi_i}\nabla_{\phi_j} J_j(\phi^*)]^\top \nabla_{\phi_j} J_i(\phi^*)$, which need not vanish in general-sum games because $\nabla_{\phi_j} J_i(\phi^*) \neq 0$ in general. Since $J_v := Dv(\phi^*)$ is invertible under SOS, the implicit function theorem gives a C^1 branch $\lambda \mapsto \phi^*_\lambda$ with $F_\lambda(\phi^*_\lambda) = 0$ and $\|\phi^*_\lambda - \phi^*\| = O(\lambda)$. Write $J_M := DM(\phi^*)$ and note that $DM(\phi^*_\lambda) = J_M + O(\lambda)$, so evaluating the shaping Jacobian at ϕ^*_λ instead of ϕ^* changes the λ-weighted linearization only at order $O(\lambda^2)$. Define*

$$\mu_M := -\lambda_{\max}\left(\tfrac{J_M + J_M^\top}{2}\right).$$

Weyl's inequality gives

$$-\lambda_{\max}\left(\tfrac{J_\lambda + J_\lambda^\top}{2}\right) \geq \mu + \lambda\mu_M - O(\lambda^2), \quad J_\lambda := DF_\lambda(\phi^*_\lambda). \tag{2}$$

If DF_λ is L_λ-Lipschitz on $\mathcal{B}_{\rho_\lambda}(\phi^*_\lambda)$, then on that ball with $\rho_\lambda := (\mu + \lambda\mu_M)/(2L_\lambda)$ (up to $O(\lambda^2)$),

$$\langle F_\lambda(\phi), \phi - \phi^*_\lambda\rangle \leq -\tfrac{1}{2}(\mu + \lambda\mu_M)\|\phi - \phi^*_\lambda\|^2.$$

The certified SOS radius around the shifted *fixed point ϕ^*_λ scales at least linearly in λ whenever $\mu_M > 0$. Under the annealed schedule (Theorem 7) the shift is transient, $\phi^*_\lambda \to \phi^*$ as $\lambda_n \to 0$, so the asymptotic target remains the base-game Nash. If $\mu_M \leq 0$, opponent-aware shaping does not enlarge the certified local basin.*

Theorem 7 (Two-phase Meta-MAPG) *Fix an SOS Nash equilibrium ϕ^* and let $r_{\text{att}} > 0$ be an attracting radius for ϕ^* under Theorem 5. Let $\bar{\lambda} > 0$ and $c_{\text{shift}} > 0$ be local constants such that the shifted fixed-point branch from Theorem 6 satisfies $\|\phi^*_\lambda - \phi^*\| \leq c_{\text{shift}}\lambda$ for all $\lambda \in [0, \bar{\lambda}]$. For a chosen $\lambda_c \in (0, \bar{\lambda}]$, let $\phi^*_{\lambda_c}$ be the shifted fixed point of $F_{\lambda_c} = v + \lambda_c M$, and let $\mathcal{B}_\rho = \{\phi : \|\phi - \phi^*_{\lambda_c}\| \leq \rho\}$ be the phase-1 SOS ball from Theorem 6 with drift constant $\mu_{\lambda_c} > 0$. Assume*

$$\rho \leq r_{\text{att}}, \qquad \lambda_c \leq \frac{\rho}{4c_{\text{shift}}}.$$

Run Meta-MAPG in two phases. In phase 1, use constant shaping $\lambda_n = \lambda_c$ and constant step size α_0 for N_0 iterations; if $\phi_{N_0} \notin \mathcal{B}_{\rho/2}$, declare the epoch failed. In phase 2, started only from $\mathcal{B}_{\rho/2}$, use $\alpha_n = \alpha_0 n^{-p}$ and $\lambda_n = \lambda_c n^{-q}$ with $1/2 < p \leq 1$ and $q > 1 - p$, together with Assumption 3.

If the phase-1 estimator has conditional variance at most σ^2 and bias norm at most β_0, then there is a constant C depending on local bounds of F_{λ_c} such that

$$\begin{aligned}&\Pr(\phi_{N_0} \notin \mathcal{B}_{\rho/2} \mid \phi_0 \in \mathcal{B}_\rho)\\ &\quad\leq \frac{4}{\rho^2}\left(V_0 e^{-\mu_{\lambda_c}\alpha_0 N_0} + C\alpha_0(\beta_0^2 + \alpha_0\sigma^2)\right),\end{aligned}$$

where $V_0 = \|\phi_0 - \phi^*_{\lambda_c}\|^2$. Hence for any target failure probability δ, choosing $N_0 = O((\mu_{\lambda_c}\alpha_0)^{-1}\log(V_0/(\delta\rho^2)))$ and α_0 small enough gives phase-2 entry with probability at least $1 - \delta$. On this event,

$$\|\phi_{N_0} - \phi^*\| \leq \|\phi_{N_0} - \phi^*_{\lambda_c}\| + \|\phi^*_{\lambda_c} - \phi^*\| \leq \frac{\rho}{2} + c_{\text{shift}}\lambda_c \leq \frac{3\rho}{4} < r_{\text{att}},$$

so phase 2 starts inside the attracting radius of Theorem 5. The annealed schedule then pulls the shifted fixed-point branch back to ϕ^*, and Theorem 5 gives $\phi_n \to \phi^*$ almost surely.

Corollary 8 (Basin-entry speedup at a common radius) *Let $d_\lambda := \|\phi^*_\lambda - \phi^*\|$ and assume $d_\lambda < \rho_0$, where ρ_0 is the SOS radius of the base field around ϕ^*. Define the common-ball radius around the* shifted *center*

$$\rho_\star := \min(\rho_\lambda, \rho_0 - d_\lambda),$$

so that $\mathcal{B}_{\rho_\star}(\phi^*_\lambda) \subseteq \mathcal{B}_{\rho_\lambda}(\phi^*_\lambda) \cap \mathcal{B}_{\rho_0}(\phi^*)$. Under the drift-improvement condition of Theorem 6, the phase-1 horizon to reach $\mathcal{B}_{\rho_\star/2}(\phi^*_\lambda)$ scales as $N_0(\lambda) = O(1/(\alpha_0\mu_\lambda))$ with a constant matched across λ (same target radius $\rho_\star$, same δ). Thus, relative to PG, Meta-MAPG shaping reduces the certified basin-entry horizon by

$$\frac{N_0(0)}{N_0(\lambda)} \gtrsim 1 + \frac{\lambda\mu_M}{\mu}$$

up to higher-order local-linearisation terms. The phase-1 Lyapunov prefactor $V_0(\lambda) = \|\phi_0 - \phi^*_\lambda\|^2$ depends on λ through the shift of the center; for matched initial offsets inside the common ball this dependence is $O(\lambda^2)$ and is absorbed into the same higher-order term. If $\mu_M \leq 0$, opponent-aware shaping need not enlarge the basin.

Theorem 9 (Restart globalisation of two-phase Meta-MAPG) *Let each restart epoch run the two-phase procedure of Theorem 7 with the same phase-1 ball $\mathcal{B}_\rho = \{\phi : \|\phi - \phi^*_{\lambda_c}\| \leq \rho\}$ on which F_{λ_c} satisfies the SOS drift condition with constant $\mu_{\lambda_c} > 0$. Let the restart distribution ν satisfy $p_\rho := \nu(\mathcal{B}_\rho) > 0$. If epochs are independent, then the process enters a successful phase-2 epoch after finitely many restarts almost surely. If T is the index of the first successful epoch, then*

$$\Pr(T > k) \leq (1 - p_\rho(1-\delta))^k, \qquad \mathbb{E}[T] \leq \frac{1}{p_\rho(1-\delta)}.$$

Thus, after finitely many restarts, the process enters an epoch where the local theorem applies and then converges almost surely to a stable Nash equilibrium in $\mathcal{N}$. This globalises a local theorem by search; it does not make the vector field globally contracting. In the worst case, if ν is roughly uniform on a radius-R d-dimensional compact set and $\mathcal{B}_\rho$ contains a radius-ρ ball, the bound scales as $O((R/\rho)^d)$. High-dimensional domains therefore need structured or warm-start restarts.

Remark 10 (Worst case vs empirics) *The $(R/\rho)^d$ bound is worst-case for uniform ν; the empirical $\mathbb{E}[T] \approx 1.24$ in Stag Hunt (Section 7) reflects a non-uniform, near-warm-start restart distribution that concentrates mass in the cooperative basin rather than any tightness of the bound.*

5. Why the Theory Works

Step 1: estimator decomposition. This is the Meta-MAPG-specific step. The theorem of Kim et al. [7] splits the sampled gradient into current, own-learning, and peer-learning score-function terms. Bounded rewards and scores give finite conditional variance; subtracting the conditional mean gives martingale noise. Finite unrolls and approximate opponent models give the bias b_n, controlled by Assumption 3.

Step 2: summability. The bias and variance terms are checked against the stochastic-approximation conditions. Because $\sum_n \alpha_n^2 < \infty$, the square-integrable martingale noise is summable in the Robbins–Siegmund sense [6, 11, 12]. Because $\sum_n \alpha_n \beta_n < \infty$, the deterministic truncation bias is also summable.

Step 3: local theorem transfer. Map the Meta-MAPG update to Giannou et al.'s local PG template. In the annealed regime, the limiting field is Nash-consistent: local stationary points are the same stable Nash policies as for the base game-gradient field. The Lyapunov recursion is the same as in Giannou et al. [5], up to summable forcing terms.

Step 4: two-phase transfer. Theorem 7 separates finite-time shaping from asymptotic convergence. Phase 1 uses constant shaping only for basin entry, with drift constant μ_{λ_c} from Theorem 6. After entry into $\mathcal{B}_{\rho/2}$, the schedule anneals; Theorem 5 follows from the local attraction theorem for stable SOS Nash policies. No global monotonicity is required. The argument is local, as in Giannou et al.

Step 5: restart globalisation. Restarts globalise the local result by a geometric argument. Each epoch is independent. The restart distribution assigns mass p_ρ to the phase-1 region, and the two-phase theorem reaches phase 2 with probability at least $1 - \delta$. Each epoch therefore succeeds with probability at least $p_\rho(1 - \delta)$. The first successful epoch is geometrically dominated. Basin maps in Section 7 show how opponent-aware shaping can improve this constant by increasing basin-entry probability.

6. Related Work

Positioning. Prior work builds opponent-shaping algorithms. LOLA differentiates through an opponent update [3]; SOS stabilises shaping in differentiable games [9]; COLA enforces consistency of mutual shaping [14]; POLA adds a proximal view [16]; M-FOS learns long-horizon shaping policies without a differentiable opponent model [10]. Our paper sits closest to Kim et al. [7]: we inherit the current/own/peer decomposition and ask a narrower question, namely when finite-unroll Meta-MAPG fits stochastic approximation, improves basin entry, and preserves the asymptotic Nash target.

Kim et al. study Meta-MAPG as a general meta-learning algorithm for rapid adaptation in mixed-incentive, competitive, and cooperative environments. Our focus is narrower and more local: we isolate the stochastic-approximation structure needed for convergence, make the basin-shaping

Table 1: Positioning among opponent-aware methods. "Nash-preserving" means the stated theory preserves the original game's stable Nash target rather than merely optimising a shaped meta-objective.

Method	Convergence guarantee	Shaping mechanism	Opponent model	Nash-preserving
LOLA	No general	1-step gradient	White-box	No
SOS	Differentiable games	Stable OS	White-box	SFP-focused
COLA	Simple-game stability	Consistent OS	White-box	No
POLA	Empirical/proximal	Proximal OS	Learned	Not primary
M-FOS	Empirical	Meta-policy OS	No	No
This work	Local SA + restarts	Two-phase Meta-MAPG	Sample estimates	Yes, by annealing

role of the peer term explicit, and use restarts as the mechanism that turns a local theorem into an equilibrium-search statement.

7. Experiments

The experiments are small by design. We use tabular Bernoulli policies in Stag Hunt and finite-horizon iterated Prisoner's Dilemma (IPD). Stag Hunt is the core test: two stable Nash equilibria, non-trivial basins, and a two-dimensional policy space where the drift constants μ and μ_M in Theorem 6 can be measured. It is also the standard testbed for LOLA, SOS, and COLA [3, 9, 14]. IPD tests the same estimator in a finite-horizon sequential game. Our Stag Hunt is a one-state, one-step, two-agent Markov game. Each agent samples C or D from a Bernoulli policy, with row-player payoff matrix

	C	D
C	4	0
D	3	2

and the column-player payoff given by the transpose. Thus (C, C) is payoff-dominant with social welfare 8, while (D, D) is risk-dominant with social welfare 4. In the figures, each coordinate is an agent's cooperation probability $p_i = \pi_i(C)$. Success means both final cooperation probabilities exceed 0.82.

Updates are sample-based: for a trajectory score $s_i = \nabla_{\phi_i} \log p_\phi(\tau)$ and return R_i, the base estimator is $\hat{g}_i = \frac{1}{B}\sum_b R_i^{(b)} s_i^{(b)}$. The own-learning term uses the likelihood-ratio Hessian estimator

$$\widehat{H}_{ii} = \frac{1}{B}\sum_b R_i^{(b)} \left(s_i^{(b)} s_i^{(b)\top} + \nabla^2_{\phi_i} \log p_\phi(\tau^{(b)}) \right),$$

and the peer-learning term uses the cross-score estimator

$$\widehat{C}_{ji} = \frac{1}{B}\sum_b R_j^{(b)} s_j^{(b)} s_i^{(b)\top}.$$

The implemented one-step correction is $\eta \widehat{H}_{ii}^\top \hat{g}_i$ for own learning and $\eta \widehat{C}_{ji}^\top \widehat{\nabla_{\phi_j} J_i}$ for peer learning. We compare PG, Meta-PG (no peer term), peer-only LOLA-style shaping (no own term), and full

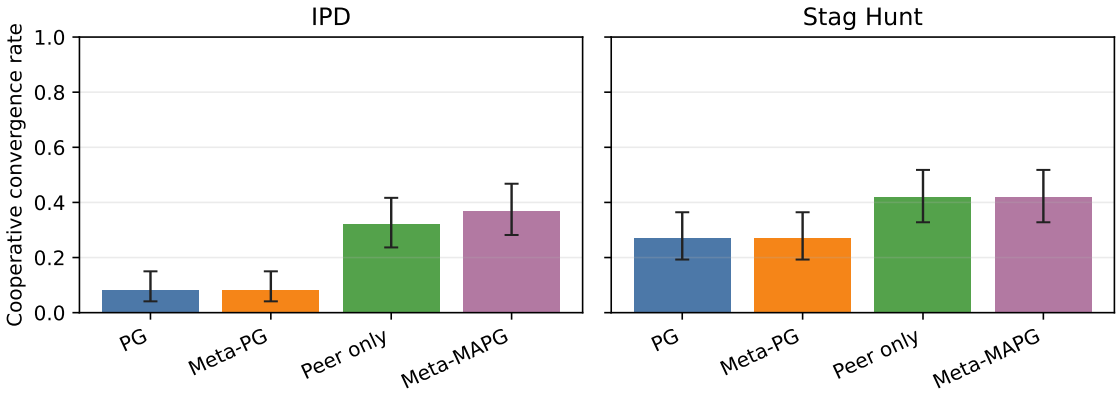

Figure 1: Component ablation with Wilson 95% confidence intervals over 100 seeds. Peer-aware methods (peer-only, Meta-MAPG) separate from non-peer methods (PG, Meta-PG) in both games. Peer-only and full Meta-MAPG are not distinguishable within CIs; the claim is the peer-learning effect, not strict dominance of the full method.

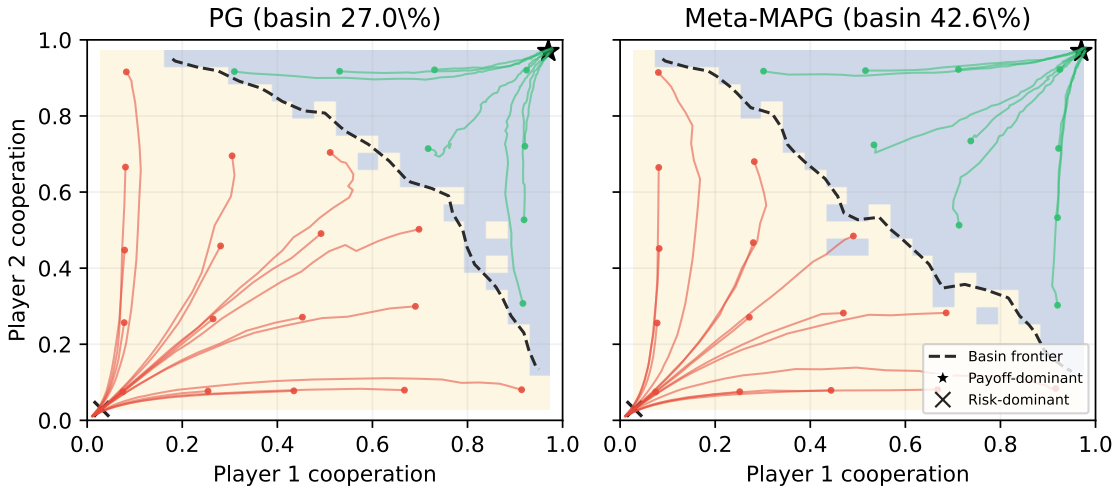

Figure 2: Basin geometry and dynamics in Stag Hunt. Background shading is the empirical basin map on a 21×21 grid (lavender = payoff-dominant, cream = risk-dominant). The dashed contour is the empirical separatrix. Overlaid trajectories start from a 5×5 sub-grid; green curves succeed, red curves fail. Meta-MAPG pushes the frontier left and down: the cooperative basin grows from 27.0% under PG to 42.6%, consistent with Theorem 6.

Meta-MAPG. Unless stated otherwise, summaries use 100 seeds; restart summaries pair initial policies across all four methods; restart-budget curves pair PG and Meta-MAPG starts; all main experiments use the same peer coefficient. Raw CSVs and run configuration are included.

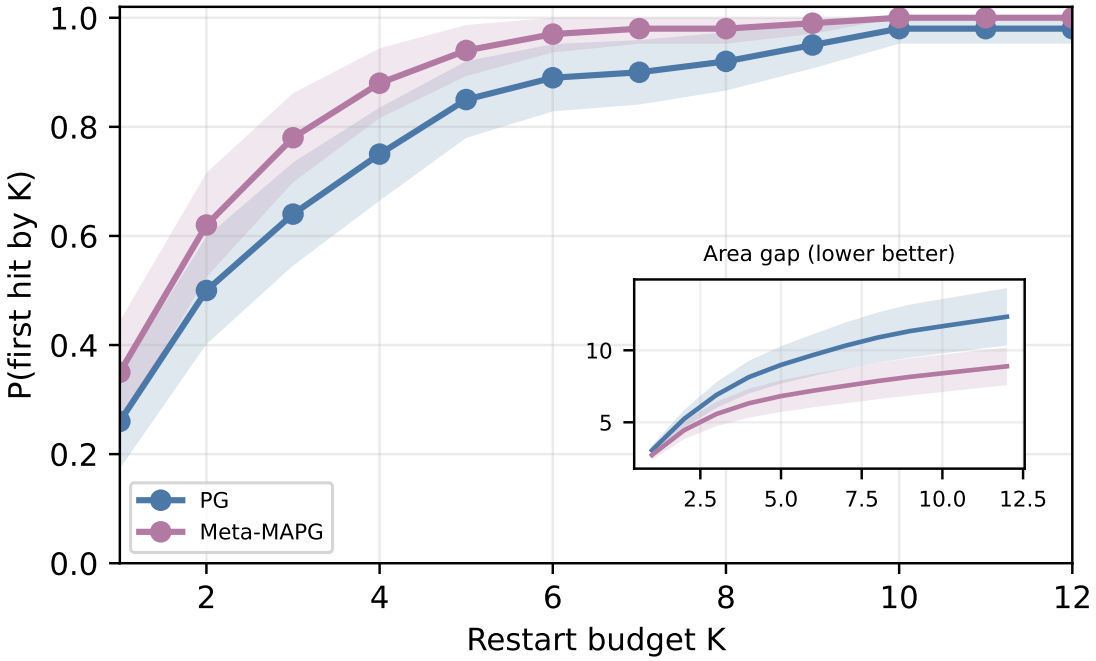

Figure 3: Restart budget as equilibrium-selection budget in Stag Hunt. With paired restart initialisations, we plot the probability of discovering the payoff-dominant equilibrium by budget K; bands show 95% confidence intervals. The inset shows $\sum_{k \leq K}(8 - W_k)$, the cumulative gap below best welfare (point estimates only, no CI band). Lower is better. Meta-MAPG reaches the good basin with fewer restarts.

The ablation shows that peer-learning terms matter; it does not show that full Meta-MAPG dominates every LOLA-style variant. Peer-only and full Meta-MAPG are indistinguishable within CIs in both Stag Hunt (42% vs 42%, 1.24 vs 1.24 restarts) and IPD (32% vs 37% within Wilson CI), so the empirical opponent-aware signal reduces to the peer term; disentangling the own-term likely requires longer unrolls (an $L \in \{1, 3, 5\}$ sweep in IPD is a natural follow-up). The trajectory figure shows the finite-time mechanism: opponent-aware shaping changes the local flow where the basin map changes. The restart-budget, basin, and peer-sweep experiments support Theorems 6 to 9: restarts turn local convergence into equilibrium search, and shaping improves that search by increasing cooperative basin mass and first-passage probability. Figure 5 runs the two-phase schedule directly. It shows that annealing after N_0 steps preserves basin-entry performance while restoring the Nash target. The MLP-policy IPD check in Appendix G is a transfer check, not a deep-MARL benchmark. The estimator sanity check in Appendix H is not a proof of Assumption 2; it checks that the sample estimator moves in the predicted direction as batch size grows.

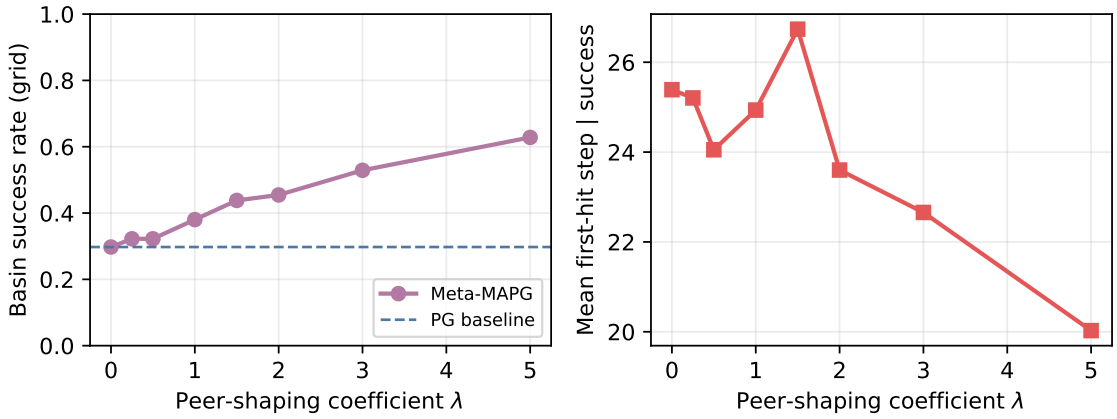

Figure 4: Peer-shaping sweep on Stag Hunt. Left: payoff-dominant basin fraction on an 11×11 initial-policy grid as peer coefficient λ varies. The rate grows from the PG baseline (29.8%) to 62.8% at $\lambda = 5$, roughly linearly for small λ, as predicted by $\rho_\lambda \gtrsim \rho_0 + \lambda\mu_M/(2L)$. Right: conditional mean first-hit step contracts from ≈ 25 to ≈ 20, matching Theorem 8.

Table 2: Main 100-seed summary. Cooperative success uses a minimum initial-state cooperation threshold of 0.82. Restart success and mean restarts are reported for all methods using paired restart initialisations.

Game	Method	Ablation success	Restart success	Restarts
stag hunt	PG	27%	100%	1.71
stag hunt	Meta-PG	27%	100%	1.75
stag hunt	Peer only	42%	100%	1.24
stag hunt	Meta-MAPG	42%	100%	1.24
ipd	PG	8%	51%	8.68
ipd	Meta-PG	8%	57%	7.90
ipd	Peer only	32%	97%	2.91
ipd	Meta-MAPG	37%	100%	2.13

8. Discussion

The theorem is local first. Meta-MAPG inherits Giannou et al.'s stable-Nash theory only inside attracting neighbourhoods and only after the estimator has stochastic-approximation form. The two-phase theorem fixes the algorithmic role: constant shaping is for basin entry; annealing restores the original Nash target. Restarts search across basins. Selecting the best endpoint makes restart

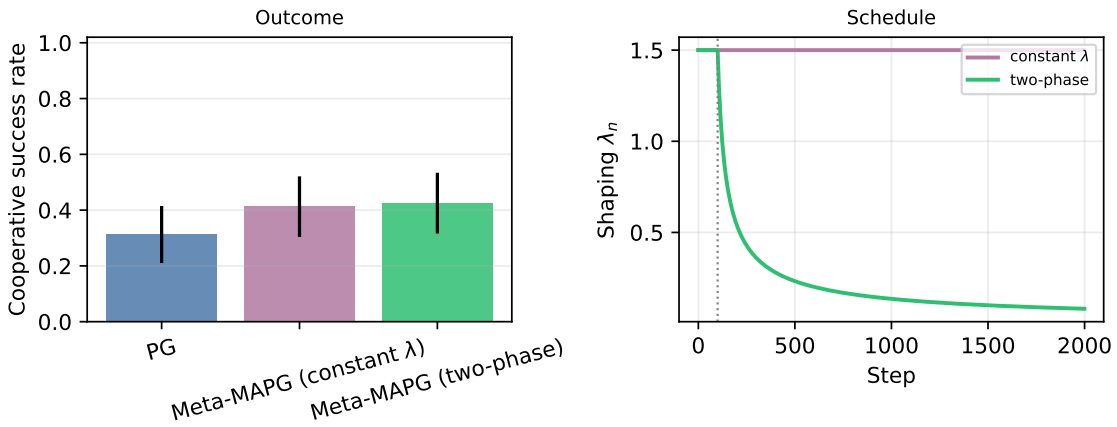

Figure 5: Two-phase schedule vs constant shaping in tabular Stag Hunt (80 seeds, 2000 steps, $\lambda_c = 1.5$). PG reaches 31.25%, constant-shaping Meta-MAPG reaches 41.25%, and the two-phase schedule reaches 42.50%; second-half cooperation std is $\leq 5 \times 10^{-4}$ in all three, so this tabular setting shows no measurable endpoint drift over the plotted horizon. The honest reading is therefore that the two-phase schedule is theoretical insurance for Nash consistency rather than an empirically separated effect in Stag Hunt. Appendix G reports the corresponding non-tabular MLP-IPD comparison. Right: λ_n over time; the dotted line marks $N_0 = 100$.

budget an equilibrium-selection resource. Its cost depends on basin probability p_ρ; shaping helps only when it increases that probability.

The assumptions are real. The estimator must have bounded conditional variance. Truncation must make unroll bias summable. Asymptotic shaping must be Nash-consistent. Constant shaping is useful for basin geometry, but a constant peer term can move fixed points in general-sum games. This is why the theorem separates annealed convergence from constant-shaping diagnostics.

The experiments are deliberately limited. Matrix and tabular games let us measure the theorem's objects: basins, restarts, drift, and estimator variance. Naive random restarts degrade with dimension: a radius-ρ basin inside a radius-R search region has worst-case mass $(\rho/R)^d$. Deep MARL needs stronger variance control, opponent models, differentiable longer unrolls, and structured restarts such as warm starts around high-welfare policies, possibly with DiCE-style estimators [4]. The claim here is narrower: opponent-aware gradients can fit stochastic approximation, and basin-entry efficiency is the right global metric when convergence is local.

We do not claim unconditional convergence of arbitrary opponent-aware learning in deep MARL. We show that finite-unroll Meta-MAPG fits classical stochastic approximation under boundedness, smoothness, Nash-consistency, and summability assumptions. Independent restarts globalise the local result when local basin mass is positive. The MLP-policy check in Appendix G only tests whether the peer-vs-nonpeer separation survives a modest neural parameterisation.

Appendix A. Proof of Theorem 4

For a sampled trajectory τ, write $S_i(\tau) = \nabla_{\phi_i} \log p_\phi(\tau)$, $H_i(\tau) = \nabla^2_{\phi_i} \log p_\phi(\tau)$, and let $R_i(\tau)$ be agent i's discounted return.

Current term. The likelihood-ratio identity [13, 15] gives the single-sample estimator $\widehat{G}_i^{\mathrm{cur}}(\tau) := R_i(\tau)S_i(\tau)$ with $\mathbb{E}[\widehat{G}_i^{\mathrm{cur}} \mid \mathcal{F}_n] = \nabla_{\phi_i} J_i(\phi_n)$.

Own-learning term. Differentiating $J_i(\phi_i + \eta\nabla_{\phi_i} J_i, \phi_{-i})$ in ϕ_i yields $\nabla_{\phi_i} J_i + \eta[\nabla^2_{\phi_i} J_i]^\top \nabla_{\phi_i} J_i$, and the Hessian identity

$$\nabla^2_{\phi_i} J_i = \mathbb{E}\Big[R_i(\tau)\Big(S_i(\tau)S_i(\tau)^\top + H_i(\tau)\Big)\Big]$$

gives the single-sample estimator $\widehat{G}_i^{\mathrm{own}}(\tau) := \eta\, \widehat{H}_{ii}(\tau)^\top \widehat{G}_i^{\mathrm{cur}}(\tau)$.

Peer-learning term. For $\phi'_j = \phi_j + \eta\nabla_{\phi_j} J_j$, differentiating $J_i(\phi_i, \phi'_j)$ in ϕ_i contributes $\eta[\nabla_{\phi_i}\nabla_{\phi_j} J_j]^\top \nabla_{\phi_j} J_i$; under per-agent parameterisation S_j has no ϕ_i-dependence, so the mixed likelihood-ratio identity collapses to

$$\nabla_{\phi_i}\nabla_{\phi_j} J_j = \mathbb{E}\Big[R_j(\tau)S_j(\tau)S_i(\tau)^\top\Big],$$

yielding $\widehat{G}_i^{\mathrm{peer}}(\tau) := \eta\, \widehat{C}_{ji}(\tau)^\top \widehat{G}_j^{\mathrm{cur}}(\tau)$.

Martingale noise and truncation bias. Each of $\widehat{G}_i^{\mathrm{cur}}, \widehat{G}_i^{\mathrm{own}}, \widehat{G}_i^{\mathrm{peer}}$ is a polynomial in the bounded-second-moment quantities R_i, R_j, S_i, S_j, H_i, hence has finite conditional second moment on $\mathcal{K}$. Setting

$$\xi^{\bullet}_{n+1} := \widehat{G}_i^{\bullet} - \mathbb{E}[\widehat{G}_i^{\bullet} \mid \mathcal{F}_n], \quad \bullet \in \{\mathrm{cur}, \mathrm{own}, \mathrm{peer}\},$$

each is a martingale difference with bounded conditional variance; so is their sum ξ_{n+1}. Longer unrolls repeat this chain-rule structure across L_n inner updates, and the truncated-tail plus approximate-opponent modelling residual is deterministic given $\mathcal{F}_n$. Collecting it into $b_n := \mathbb{E}[g_n \mid \mathcal{F}_n] - F(\phi_n)$, the bounded-derivative hypotheses on $\mathcal{K}$ and Assumption 3 give $\|b_n\| \le C\beta_n$ with $\beta_n \to 0$. Assembling the three components produces

$$g_n = F(\phi_n) + b_n + \xi_{n+1},$$

which is exactly Assumption 2. □

Appendix B. Local Lyapunov Drift

We spell out the stochastic-approximation step behind Theorem 5. The goal is not to reprove Giannou et al. [5]; it is to show that Meta-MAPG has the same Lyapunov form after decomposition.

Lemma 11 (One-step local Lyapunov drift) *Fix an SOS Nash equilibrium ϕ^* and a ball $\mathcal{B}_\rho = \{\phi : \|\phi - \phi^*\| \le \rho\}$ on which the effective field F satisfies*

$$\langle F(\phi), \phi - \phi^* \rangle \le -\mu\|\phi - \phi^*\|^2.$$

Suppose the stopped iterates remain in $\mathcal{B}_\rho$ and the Meta-MAPG update admits

$$g_n = F(\phi_n) + \xi_{n+1} + b_n,$$
$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0, \qquad \mathbb{E}[\|\xi_{n+1}\|^2 \mid \mathcal{F}_n] \le \sigma^2,$$

with $\|b_n\| \le \beta_n$. *If* $V_n = \|\phi_n - \phi^*\|^2$ *and* $\|F(\phi)\| \le C_F$ *on* $\mathcal{B}_\rho$, *then*

$$\begin{aligned}\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \le (1 - 2\mu\alpha_n)V_n + 2\alpha_n\beta_n\sqrt{V_n} \\ + 3\alpha_n^2(C_F^2 + \beta_n^2 + \sigma^2).\end{aligned} \tag{3}$$

Proof Projection onto the convex compact set $\mathcal{K}$ is non-expansive, so it is enough to bound the unprojected step. Expanding the square gives

$$\begin{aligned}V_{n+1} &\le \|\phi_n - \phi^* + \alpha_n g_n\|^2 \\ &= V_n + 2\alpha_n\langle g_n, \phi_n - \phi^*\rangle + \alpha_n^2\|g_n\|^2.\end{aligned}$$

Taking conditional expectation and using $\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0$,

$$\begin{aligned}\mathbb{E}[\langle g_n, \phi_n - \phi^*\rangle \mid \mathcal{F}_n] &= \langle F(\phi_n), \phi_n - \phi^*\rangle \\ &\quad + \langle b_n, \phi_n - \phi^*\rangle \\ &\le -\mu V_n + \beta_n\sqrt{V_n}.\end{aligned}$$

For the quadratic term, $(a + b + c)^2 \le 3(a^2 + b^2 + c^2)$ gives

$$\begin{aligned}\mathbb{E}[\|g_n\|^2 \mid \mathcal{F}_n] &\le 3\big(\|F(\phi_n)\|^2 + \|b_n\|^2 \\ &\quad + \mathbb{E}[\|\xi_{n+1}\|^2 \mid \mathcal{F}_n]\big) \\ &\le 3(C_F^2 + \beta_n^2 + \sigma^2).\end{aligned}$$

Substituting the last two displays into the square expansion proves (3). ■

If $\sum_n \alpha_n\beta_n < \infty$ and $\sum_n \alpha_n^2 < \infty$, then (3) is in Robbins–Siegmund form. Indeed,

$$2\alpha_n\beta_n\sqrt{V_n} \le \alpha_n\beta_n(1 + V_n),$$

so for all large n,

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \le (1 - a_n)V_n + \eta_n$$

with $\sum_n a_n = \infty$ and $\sum_n \eta_n < \infty$. Robbins–Siegmund yields $V_n \to V_\infty$ almost surely and $\sum_n a_n V_n < \infty$; since $\sum_n a_n = \infty$, $V_\infty = 0$. Thus $\phi_n \to \phi^*$ almost surely on the local event.

Appendix C. Proof of Proposition 6

We prove the three parts separately.

Fixed-point shift. $F_\lambda(\phi^*) = \lambda M(\phi^*)$, where $M(\phi^*) = M^{\text{peer}}(\phi^*) \ne 0$ in general (own-learning vanishes because $\nabla_{\phi_i} J_i(\phi^*) = 0$). Since $J_v = Dv(\phi^*)$ is non-singular under SOS, the implicit function theorem applied to $\Phi(\lambda, \phi) := F_\lambda(\phi)$ at $(0, \phi^*)$ gives a C^1 branch ϕ_λ^* with $F_\lambda(\phi_\lambda^*) = 0$ and

$$\phi_\lambda^* = \phi^* - \lambda J_v^{-1} M(\phi^*) + O(\lambda^2),$$

so $\|\phi_\lambda^* - \phi^*\| = O(\lambda)$ for small λ.

Linearised drift bound. Both J_v and J_M are real matrices; their symmetric parts $\mathrm{sym}(A) := (A + A^\top)/2$ are Hermitian. By definition $-\lambda_{\max}(\mathrm{sym}(J_v)) = \mu$ and $-\lambda_{\max}(\mathrm{sym}(J_M)) = \mu_M$. Because $DM(\phi^*_\lambda) = DM(\phi^*) + O(\|\phi^*_\lambda - \phi^*\|) = J_M + O(\lambda)$, the shaping Jacobian evaluated at the shifted fixed point differs from J_M by $O(\lambda)$, which becomes an $O(\lambda^2)$ contribution after multiplying by λ. Thus $DF_\lambda(\phi^*_\lambda) = J_v + \lambda J_M + O(\lambda^2)$. Weyl's inequality gives

$$\lambda_{\max}\big(\mathrm{sym}(J_v + \lambda J_M)\big) \le \lambda_{\max}\big(\mathrm{sym}(J_v)\big) + \lambda\, \lambda_{\max}\big(\mathrm{sym}(J_M)\big),$$

and negating, together with the $O(\lambda^2)$ evaluation shift, yields (2).

Certified radius. Let $\mu_\lambda := \mu + \lambda\mu_M$. Fix $\phi \in \mathcal{B}_{\rho_\lambda}(\phi^*_\lambda)$ and write $x := \phi - \phi^*_\lambda$. Since $F_\lambda(\phi^*_\lambda) = 0$, Taylor's theorem with integral remainder gives

$$F_\lambda(\phi) = J_\lambda x + R_\lambda(x), \qquad \|R_\lambda(x)\| \le \tfrac{1}{2} L_\lambda \|x\|^2,$$

with $J_\lambda = DF_\lambda(\phi^*_\lambda)$ and L_λ the Lipschitz constant of DF_λ on $\mathcal{B}_{\rho_\lambda}(\phi^*_\lambda)$. Taking inner products,

$$\langle J_\lambda x, x\rangle = x^\top \mathrm{sym}(J_\lambda) x \le -\mu_\lambda \|x\|^2$$

by Rayleigh and (2) (absorbing the $O(\lambda^2)$ term into μ_λ for small λ), and $|\langle R_\lambda(x), x\rangle| \le \frac{1}{2} L_\lambda \|x\|^3$. Hence

$$\langle F_\lambda(\phi), \phi - \phi^*_\lambda\rangle \le -\mu_\lambda \|x\|^2 + \tfrac{1}{2} L_\lambda \|x\|^3.$$

For $\|x\| \le \rho_\lambda := \mu_\lambda/(2L_\lambda)$, the cubic term is at most $(\mu_\lambda/4)\|x\|^2$, so

$$\langle F_\lambda(\phi), \phi - \phi^*_\lambda\rangle \le -\tfrac{1}{2}\mu_\lambda \|x\|^2,$$

the stated SOS drift on $\mathcal{B}_{\rho_\lambda}(\phi^*_\lambda)$. □

Appendix D. Proof of Theorem 7

Write $V_n = \|\phi_n - \phi^*_{\lambda_c}\|^2$ (phase-1 Lyapunov, centred at the shifted fixed point of Theorem 6) and let $\tau := \inf\{n \le N_0 : \phi_n \notin \mathcal{B}_\rho\}$ with $\tau = N_0 + 1$ if the iterates never leave. On the event $\{n < \tau\}$ the phase-1 drift of Theorem 6 applies with constant μ_{λ_c}.

Step 1: one-step drift. On $\{n < \tau\}$, by Assumption 2 and the phase-1 drift bound around $\phi^*_{\lambda_c}$,

$$\begin{aligned}
\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}\big[\|\phi_n - \phi^*_{\lambda_c} + \alpha_0 g_n\|^2 \mid \mathcal{F}_n\big] \\
&= V_n + 2\alpha_0 \langle F_{\lambda_c}(\phi_n), \phi_n - \phi^*_{\lambda_c}\rangle \\
&\quad + 2\alpha_0 \langle b_n, \phi_n - \phi^*_{\lambda_c}\rangle + \alpha_0^2 \mathbb{E}[\|g_n\|^2 \mid \mathcal{F}_n] \\
&\le V_n - 2\alpha_0 \mu_{\lambda_c} V_n + 2\alpha_0 \beta_0 \sqrt{V_n} + \alpha_0^2 G^2,
\end{aligned}$$

where $G^2 := 3(C^2_{F_{\lambda_c}} + \beta_0^2 + \sigma^2)$ is a uniform bound on the conditional second moment of g_n (with $C_{F_{\lambda_c}}$ the sup-norm of F_{λ_c} on $\mathcal{B}_\rho$). Applying $2ab \le \eta a^2 + \eta^{-1} b^2$ with $a = \sqrt{V_n}$, $b = \beta_0$, and $\eta = \mu_{\lambda_c}$,

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \le (1 - \mu_{\lambda_c}\alpha_0) V_n + \frac{\alpha_0 \beta_0^2}{\mu_{\lambda_c}} + \alpha_0^2 G^2.$$

Step 2: unconditional recursion on the stopped process. Define the stopped sequence $\tilde{V}_n := V_{n\wedge\tau}$. The drift bound and the fact that $\tilde{V}_{n+1} = \tilde{V}_n$ on $\{n \geq \tau\}$ give

$$\mathbb{E}[\tilde{V}_{n+1}] \leq (1 - \mu_{\lambda_c}\alpha_0)\mathbb{E}[\tilde{V}_n] + c_1,$$
$$c_1 := \frac{\alpha_0\beta_0^2}{\mu_{\lambda_c}} + \alpha_0^2 G^2.$$

Iterating,

$$\mathbb{E}[\tilde{V}_{N_0}] \leq (1 - \mu_{\lambda_c}\alpha_0)^{N_0} V_0 + \frac{c_1}{\mu_{\lambda_c}\alpha_0}.$$

Using $(1-x)^k \leq e^{-xk}$ for $x \in [0,1]$,

$$\mathbb{E}[\tilde{V}_{N_0}] \leq V_0\, e^{-\mu_{\lambda_c}\alpha_0 N_0} + \frac{\beta_0^2}{\mu_{\lambda_c}^2} + \frac{\alpha_0 G^2}{\mu_{\lambda_c}}.$$

Step 3: Markov's inequality. If $\phi_{N_0} \notin \mathcal{B}_{\rho/2}$ and $\tau > N_0$, then $V_{N_0} \geq (\rho/2)^2$. If instead $\tau \leq N_0$, then by definition of τ we have $\phi_\tau \notin \mathcal{B}_\rho$, so $V_\tau > \rho^2 \geq (\rho/2)^2$, and the stopped process freezes at $\tilde{V}_{N_0} = V_\tau$. Hence in both cases $\tilde{V}_{N_0} \geq (\rho/2)^2$. Therefore

$$\begin{aligned}&\Pr(\phi_{N_0} \notin \mathcal{B}_{\rho/2} \mid \phi_0 \in \mathcal{B}_\rho)\\ &\quad\leq \Pr(\tilde{V}_{N_0} \geq (\rho/2)^2) \leq \frac{4\mathbb{E}[\tilde{V}_{N_0}]}{\rho^2}.\end{aligned}$$

Combining with Step 2,

$$\begin{aligned}&\Pr(\phi_{N_0} \notin \mathcal{B}_{\rho/2} \mid \phi_0 \in \mathcal{B}_\rho)\\ &\quad\leq \frac{4}{\rho^2}\left(V_0 e^{-\mu_{\lambda_c}\alpha_0 N_0} + C\alpha_0(\beta_0^2 + \alpha_0\sigma^2)\right),\end{aligned}$$

for a constant C depending on $\mu_{\lambda_c}, C_{F_{\lambda_c}}$.

Step 4: choice of N_0. Given a target $\delta \in (0,1)$, set

$$N_0 = \left\lceil \frac{1}{\mu_{\lambda_c}\alpha_0} \log\frac{8V_0}{\delta\rho^2} \right\rceil, \qquad \alpha_0 \leq \frac{\delta\rho^2}{8C(\beta_0^2 + \sigma^2)}.$$

Each summand in Step 3 is then at most $\delta/2$, so the total failure probability is at most δ.

Step 5: phase 2. Conditional on $\phi_{N_0} \in \mathcal{B}_{\rho/2}$, the subsequent recursion uses $\alpha_n = \alpha_0 n^{-p}$ and $\lambda_n = \lambda_c n^{-q}$ with $1/2 < p \leq 1$ and $q > 1 - p$, so $\sum_n \alpha_n = \infty$, $\sum_n \alpha_n^2 < \infty$, and $\sum_n \alpha_n\lambda_n < \infty$ by direct integral comparison. By the triangle inequality and the explicit shift bound in Theorem 7,

$$\|\phi_{N_0} - \phi^*\| \leq \|\phi_{N_0} - \phi^*_{\lambda_c}\| + \|\phi^*_{\lambda_c} - \phi^*\| \leq \frac{\rho}{2} + c_{\text{shift}}\lambda_c \leq \frac{3\rho}{4} < r_{\text{att}}.$$

Hence the phase-2 iterate starts inside the attracting radius of Theorem 5. The annealed fixed-point branch satisfies $\phi^*_{\lambda_n} \to \phi^*$ by Theorem 6; together with Assumptions 2, 3 and 5, this places the recursion inside the hypotheses of Theorem 5 on the event $\{\phi_{N_0} \in \mathcal{B}_{\rho/2}\}$. Conditional on this event, $\phi_n \to \phi^*$ almost surely. The total probability that the two-phase procedure converges in a single epoch is therefore at least $(1-\delta)\cdot 1 = 1 - \delta$. □

Appendix E. Proof of Theorem 9

Let $\mathcal{F}_k^{\text{epoch}}$ denote the σ-algebra generated by epochs $1, \ldots, k-1$ (outcomes only; restart draws are independent by assumption). Let A_k be the event that epoch k starts in $\mathcal{B}_\rho$ and that the two-phase procedure of Theorem 7 reaches phase 2 and converges within that epoch.

Step 1: per-epoch lower bound. Restart k samples $\phi_0^{(k)} \sim \nu$ independently of $\mathcal{F}_k^{\text{epoch}}$, so

$$\Pr(\phi_0^{(k)} \in \mathcal{B}_\rho \mid \mathcal{F}_k^{\text{epoch}}) = \nu(\mathcal{B}_\rho) = p_\rho.$$

Conditional on $\phi_0^{(k)} \in \mathcal{B}_\rho$, Theorem 7 gives $\Pr(\text{epoch succeeds}) \geq 1 - \delta$. Therefore

$$\Pr(A_k \mid \mathcal{F}_k^{\text{epoch}}) \geq p_\rho(1-\delta) =: p_\star > 0.$$

Step 2: geometric tail. Let $T := \inf\{k \geq 1 : A_k \text{ occurs}\}$. Then

$$\begin{aligned}\Pr(T > k) &= \mathbb{E}\left[\prod_{j=1}^{k} \mathbf{1}_{A_j^c}\right] \\ &= \mathbb{E}\left[\mathbb{E}[\mathbf{1}_{A_k^c} \mid \mathcal{F}_k^{\text{epoch}}] \prod_{j=1}^{k-1} \mathbf{1}_{A_j^c}\right].\end{aligned}$$

By Step 1, $\mathbb{E}[\mathbf{1}_{A_k^c} \mid \mathcal{F}_k^{\text{epoch}}] \leq 1 - p_\star$ almost surely, so iterating,

$$\Pr(T > k) \leq (1 - p_\star)^k.$$

Hence $\Pr(T < \infty) = 1$ and

$$\mathbb{E}[T] = \sum_{k \geq 0} \Pr(T > k) \leq \sum_{k \geq 0} (1 - p_\star)^k = \frac{1}{p_\star}.$$

Step 3: dimensional dependence. If ν is lower-bounded by density $m > 0$ on a neighbourhood U_{λ_c} containing $\mathcal{B}_\rho(\phi^*_{\lambda_c})$ in $\mathbb{R}^d$, then $p_\rho \geq m\, c_d \rho^d$ where $c_d = \pi^{d/2}/\Gamma(d/2+1)$ is the volume of the unit d-ball. Because $\|\phi^*_{\lambda_c} - \phi^*\| = O(\lambda_c)$, for sufficiently small λ_c this is the same local region as a neighbourhood of ϕ^* up to shrinking U_{λ_c} by an $O(\lambda_c)$ margin. Combined with the certified radius $\rho_\lambda \geq (\mu + \lambda\mu_M)/(2L_\lambda)$ from Theorem 6, the expected number of epochs obeys

$$\mathbb{E}[T] \leq \frac{1}{(1-\delta) m c_d} \left(\frac{2L_\lambda}{\mu + \lambda\mu_M}\right)^d.$$

Shaping reduces the base of this dimensional factor whenever $\mu_M > 0$; the overall worst-case dependence is still exponential in d, as discussed in Section 4. □

Appendix F. Experiment Reproducibility

The experiment scripts, figures, CSVs, and run configurations are included. The tabular script saves its command-line configuration and outputs under `artifacts/main`. These files cover the ablation, restart, trajectory, basin, peer-sweep, annealing, and estimator-sanity figures. Updates use no exact gradients; expected returns are computed only for logging and evaluation.

The optional MLP-policy IPD check in Appendix G requires PyTorch and uses DiCE-style surrogate losses for the higher-order terms. Its aggregate ablation and annealing outputs are stored under `artifacts/mlp`, including `mlp_ipd.pdf`, `mlp_annealing.pdf`, and their per-seed summary CSVs. The repository README contains the full commands for regenerating both sets of experiments.

Appendix G. Non-Tabular IPD Check

To check that the ablation is not purely tabular, we repeat IPD with a two-layer MLP policy for each agent: a 5-dimensional one-hot previous-action input, two tanh hidden layers of width 16, and one Bernoulli cooperation logit. This scales the policy from 5 tabular logits to about 385 parameters per agent. We keep the tabular budget, batch size, seed schedule, peer coefficient, and own coefficient. Higher-order terms use PyTorch autograd with DiCE-style surrogate losses [4].

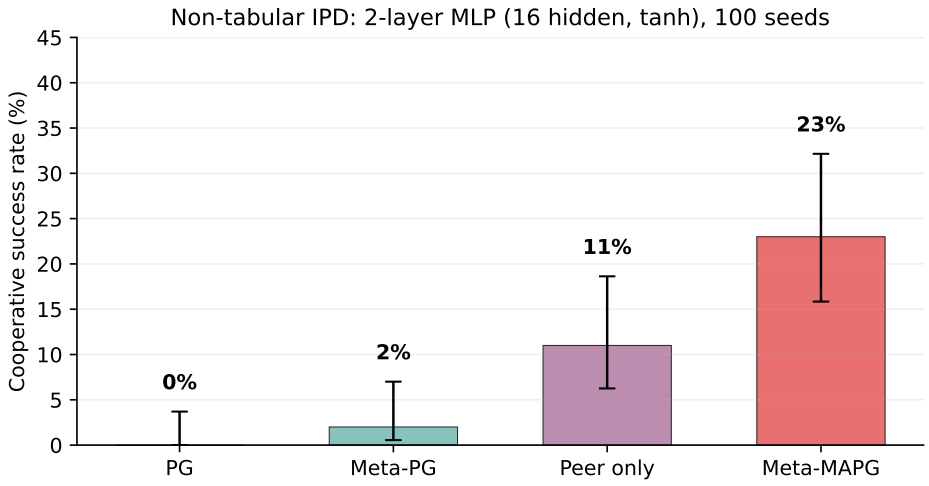

Figure 6: Non-tabular IPD check with MLP policies. Cooperative success rate over 100 seeds with Wilson 95% confidence intervals. Peer-aware methods remain above non-peer methods under the same coefficients as the tabular experiments: PG reaches 0%, Meta-PG 2%, peer-only 11%, and Meta-MAPG 23%. This is a transfer check, not a deep-MARL benchmark.

We also mirror the annealing comparison in the same MLP IPD setting. The point is not to claim a large non-tabular gain from the schedule; it is to check that the two-phase construction does not collapse once the policy is no longer tabular.

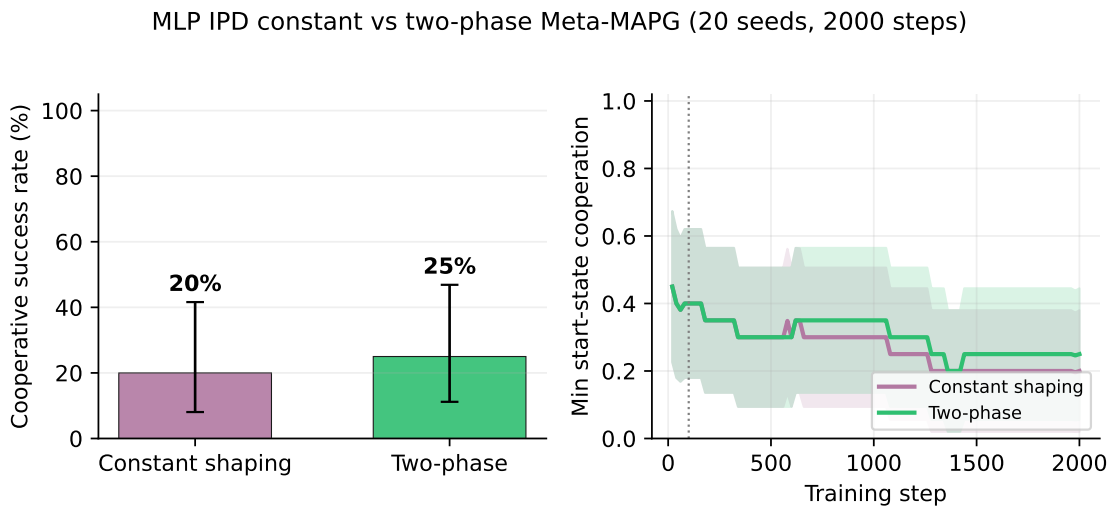

Figure 7: Constant shaping vs two-phase Meta-MAPG in MLP IPD (20 seeds, 2000 steps, $N_0 = 100$). Left: cooperative success rate with Wilson 95% confidence intervals. Constant shaping reaches 20% and two-phase reaches 25%, so the separation is small but directionally aligned with the tabular story. Right: mean minimum start-state cooperation over training, with the dotted line marking the end of phase 1. The non-tabular comparison therefore supports the narrower claim needed by Theorem 7: annealing does not materially harm basin-entry performance, even when the policy class is an MLP.

Appendix H. Estimator Sanity Check

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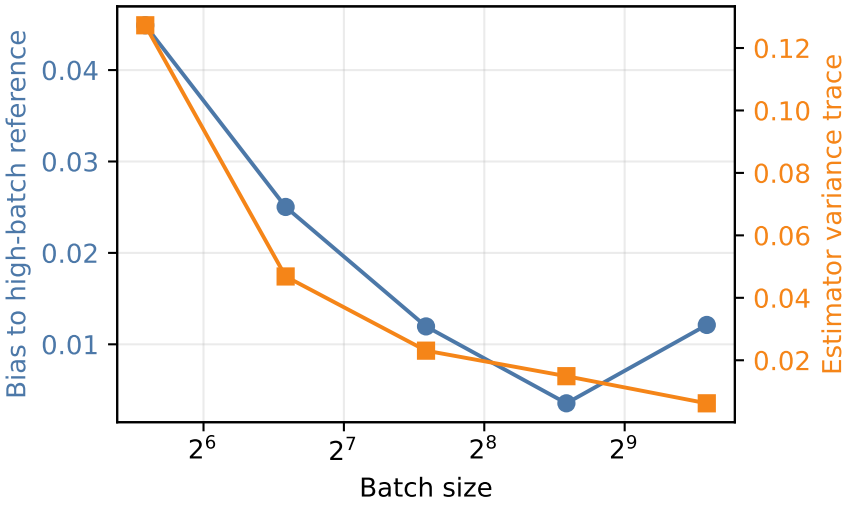

Figure 8: Estimator sanity check. At a fixed Stag Hunt parameter, the variance trace of the Meta-MAPG estimator decreases from 0.1272 at batch size 48 to 0.0062 at batch size 768. Bias is measured relative to a high-batch reference estimator, not an exact analytic gradient.

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